

- RAG Learning

1.3 Computer Networks, Connections and Protocols

OCR GCSE Computer Science (J277) — Paper 1

1.3.1 Network Types and Performance

A **computer network** is two or more computing devices connected together for the purpose of sharing data, resources, and communication. Networks allow users to share files, printers, internet access, and software licences, and enable communication via email, messaging, and video calls.

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Coverage	Small geographic area — a single building, home, school campus, or office	Large geographic area — cities, countries, or the entire world
Infrastructure ownership	All hardware (cables, switches, routers) owned and managed by the organisation	Uses third-party communication links leased from telecoms providers (e.g., BT, Virgin Media). Undersea fibre optic cables, satellite links.
Connection type	Ethernet cables (Cat5e/Cat6) or Wi-Fi	Leased lines, fibre optics, satellite, microwave links, submarine cables
Speed	Typically fast (100 Mbps to 10 Gbps)	Variable; often slower than LAN due to distance and shared infrastructure

Security	Easier to secure (physical access control, single admin, limited entry points)	Harder to secure (data travels over public infrastructure, more attack vectors)
Example	School network, home Wi-Fi, office LAN	The internet (the world's largest WAN), a bank connecting branches across the country

Exam Tip: The key distinction between LAN and WAN is **ownership of infrastructure**, not just size. A LAN uses infrastructure owned by the organisation; a WAN leases third-party infrastructure. The internet is the world's largest WAN — it connects millions of LANs together using leased and shared infrastructure.

Factors Affecting Network Performance

Network performance depends on several factors that determine how fast data can be transferred and how responsive the network feels:

- **Bandwidth** — the maximum data transfer capacity of the connection (measured in Mbps or Gbps). Higher bandwidth = more data per second. Bandwidth is the theoretical maximum; actual throughput is usually lower due to overhead and congestion. Think of bandwidth as the width of a pipe: a wider pipe can carry more water per second.
- **Number of connected devices** — more devices sharing bandwidth means less available for each. A network with 100 users will be slower per user than one with 10, because bandwidth is shared. This is especially noticeable on wireless networks where all devices share the same radio channel.
- **Latency** — the delay between sending data and it arriving at its destination. Affected by distance, number of router hops, congestion, and processing delays at each node. Measured in milliseconds (ms). Low latency is critical for gaming, video calls, and remote surgery. Even with high bandwidth, high latency makes a connection feel sluggish.
- **Wired vs wireless** — wired connections (Ethernet) are generally faster, more reliable, lower latency, and more secure than wireless (Wi-Fi). Wired connections also do not suffer from interference.

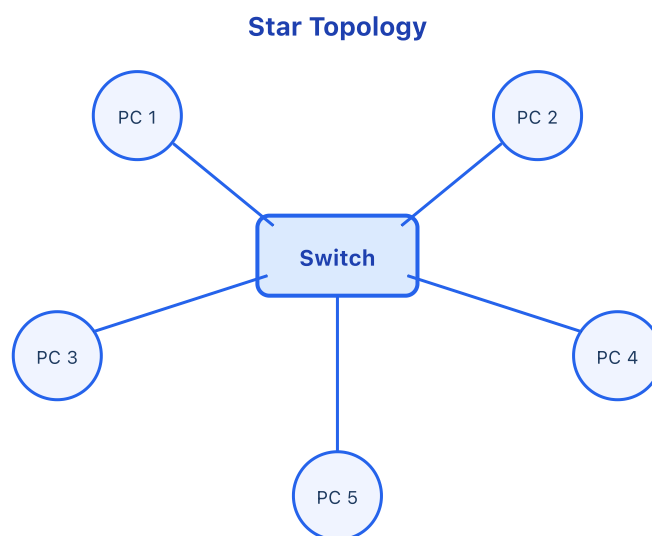
- **Interference / obstacles** — wireless signals are weakened by walls, floors, metal objects, and electromagnetic interference from other devices (microwaves, Bluetooth, neighbouring Wi-Fi networks). The further from the access point, the weaker the signal and the slower the connection.
- **Network hardware quality** — older switches, routers, or cables can bottleneck performance. A Cat5 cable limits you to 100 Mbps even if your internet connection is 1 Gbps. Upgrading to Cat6 cables and modern switches can dramatically improve LAN performance.
- **Network congestion** — during peak times (e.g., evenings when everyone is streaming video), shared infrastructure becomes congested, reducing speed for everyone.

1.3.2 Network Topologies

A **network topology** is the arrangement of devices (nodes) and connections (links) in a network. It determines how data flows and affects performance, reliability, cost, and ease of maintenance. There are two types: **physical topology** (how devices are physically connected) and **logical topology** (how data actually flows).

Star Topology

All devices connect to a **central switch** (or hub) via individual cables. The most common LAN topology in homes, schools, and businesses.

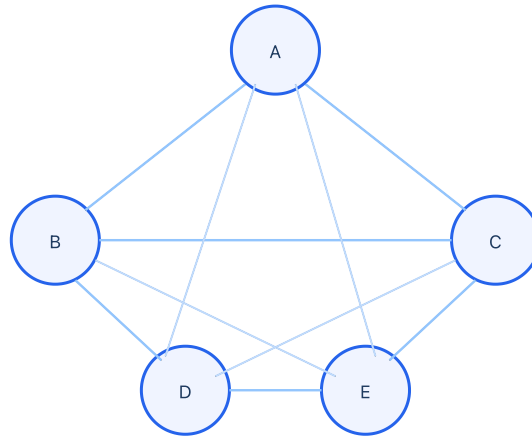


Advantages	Disadvantages
If one cable fails, only that device is affected — the rest of the network continues working normally	The central switch is a single point of failure — if it fails, the entire network goes down (mitigated by having a backup switch)
Easy to add new devices without disrupting the network (just plug into a spare switch port)	Requires more cabling than a bus topology (individual cable per device), increasing installation cost
Easy to troubleshoot — faults can be isolated to individual cables or devices quickly	The switch itself can be expensive , especially enterprise-grade managed switches
No collisions when using a switch (each port is a separate collision domain), so consistent performance	Limited by the number of ports on the switch (though switches can be daisy-chained)
High performance — each device has a dedicated connection to the switch	
Secure — switch sends data only to the intended recipient, not broadcast to all	

Mesh Topology

Full mesh: every device connects directly to every other device. **Partial mesh:** some (but not all) devices have redundant connections. The internet's core infrastructure uses a partial mesh.

Full Mesh Topology

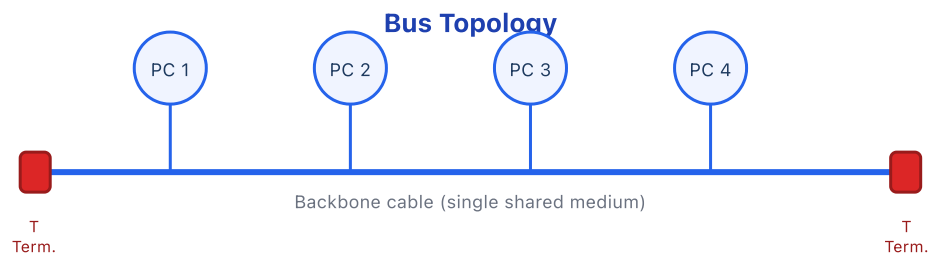


Advantages	Disadvantages
Very resilient — if one connection fails, data can be rerouted via alternative paths. No single point of failure.	Very expensive — requires a large amount of cabling. Full mesh of n devices needs $n(n-1)/2$ connections (e.g., 10 devices need 45 connections).
Multiple data paths — data can take the fastest or least congested route, improving performance	Complex to set up, configure, and maintain. Each device needs multiple network interfaces.
Fault tolerant — ideal for critical infrastructure where downtime is unacceptable	Impractical for large networks due to the quadratic growth in connections required

The **internet** uses a partial mesh topology — core routers have multiple redundant connections to ensure reliability, but not every device connects to every other. This balances cost with resilience.

Bus Topology

All devices connect to a single central cable called the **backbone** or bus. Data sent by any device travels in both directions along the backbone and is seen by all devices. Each device checks whether the data is addressed to it; if not, it ignores it. Terminators at each end of the bus absorb signals and prevent them bouncing back.

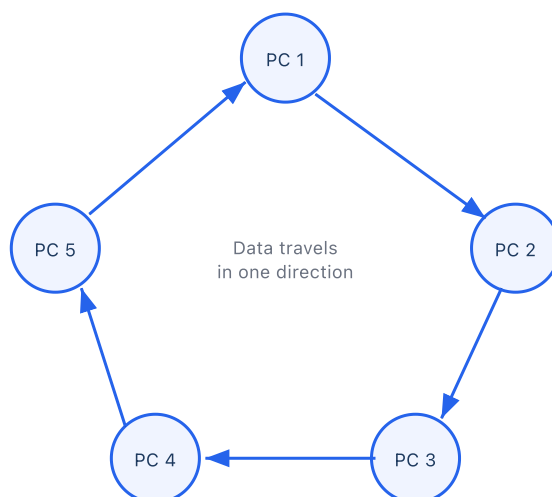


Advantages	Disadvantages
Simple and cheap to install — less cabling than star (no central switch needed, no individual cable per device)	Single point of failure — if the backbone cable breaks or a terminator fails, the <i>entire</i> network goes down
Easy to extend — add new devices by connecting to any point on the backbone	Collisions — only one device can transmit at a time. If two devices send simultaneously, their signals collide, both are corrupted, and both must retransmit after a random delay. Performance degrades badly as more devices are added.
Suitable for small networks where cost is a priority	All devices see all data — less secure (any device can sniff traffic from any other)
	Difficult to fault find — a break anywhere in the backbone kills everything; isolating the fault is hard
	Now largely obsolete — replaced by star topology in all modern LANs

Ring Topology

Each device connects to exactly two others, forming a closed loop. Data travels around the ring in one direction (or both directions in a dual-ring), passing through each device in sequence until it reaches its destination.

Ring Topology



Advantages	Disadvantages
Data travels in one direction — no collisions , unlike a bus topology. Each device acts as a repeater, boosting the signal before passing it on.	If any device or cable fails , the entire ring is broken and the whole network goes down (unless a dual-ring or bypass mechanism is used)
Performs well under heavy load — token-ring protocols ensure only one device transmits at a time without collisions	Adding or removing a device disrupts the whole network — the ring must be broken to insert or remove a node
Equal access for all devices — no device can monopolise the network	Difficult to fault find — a problem anywhere on the ring affects everything
	Now mostly obsolete for wired LANs (replaced by star). Still used in some WAN and fibre metropolitan networks (dual-ring SONET/SDH).

Topology Comparison Summary

Feature	Star	Bus	Ring	Mesh
Single point of failure	Switch (but devices still	Backbone cable or	Any device or cable	None (full mesh)

	isolated)	terminator		
Collisions?	No (switch)	Yes (shared medium)	No (token-based)	No (direct paths)
Cabling cost	Medium (one cable per device)	Low (single backbone)	Medium (each device connects to two others)	Very high ($n(n-1)/2$ cables)
Fault tolerance	Medium	Very low	Low	Very high
Common today?	Yes — standard LAN	Obsolete	Mostly obsolete	Internet backbone, critical infrastructure

Exam Tip: The star topology is the most important for GCSE — know it in detail. For bus topology, the key weaknesses are: (1) single backbone = single point of failure for the whole network, and (2) all devices share the medium so collisions occur. For ring topology, the key weakness is that any break in the ring brings down the entire network. Mesh topology's key strength is redundancy — multiple paths mean no single point of failure. Always relate topology choice to the scenario: a school network uses star; a critical military network uses mesh.

1.3.3 Client-Server and Peer-to-Peer

These are two different models for how devices on a network interact and share resources:

Feature	Client-Server	Peer-to-Peer (P2P)
Structure	Dedicated server(s) provide services (files, web pages, email, authentication, databases); clients make requests. Clear separation of roles.	All devices (peers) are equal — each can act as both client and server simultaneously. No central authority.

Security	Centralised management — server enforces user accounts, permissions, passwords, encryption, and security policies consistently	Each peer manages its own security — harder to enforce consistent policies across all devices
Backups	Centralised backups on the server — systematic, reliable, and managed by an administrator	Each peer must back up its own data — inconsistent and unreliable
Cost	Expensive — requires dedicated server hardware, server software licences, and a network administrator to manage the system	Cheap — no dedicated hardware needed; uses existing devices. No specialist admin required.
Reliability	Server is a single point of failure (mitigated by redundant servers, RAID storage, and failover systems)	No single point of failure — network continues if individual peers go offline
Scalability	Limited by server capacity (can upgrade hardware or add more servers, but at cost)	Scales naturally — each new peer adds both demand AND capacity to the network
Examples	School/corporate networks, web browsing (client browser requests pages from web server), email systems, online gaming servers, streaming services	Home networks, BitTorrent file sharing, some VoIP (e.g., older Skype), cryptocurrency networks (Bitcoin)

Exam Tip: For client-server vs P2P, focus on the trade-offs: client-server is centralised, easier to manage and secure, but expensive and has a single point of failure. P2P is decentralised, cheaper, and more resilient, but harder to secure and manage. When given a scenario, recommend one and justify: "A school should use client-server because centralised user accounts allow the administrator to control access, enforce password policies, and manage backups from one location."

1.3.4 Network Hardware

Several hardware devices work together to enable network communication:

Device	Purpose	Key Detail
Router	Connects different networks together; routes packets between networks using IP addresses	Connects your LAN to the internet (your ISP's WAN). Examines the destination IP address in each packet and uses a routing table to determine the best next hop. A home "router" typically combines a router, switch, WAP, firewall, and DHCP server in one box. Operates at Layer 3 (Internet/Network layer).
Switch	Connects devices within a LAN ; forwards data frames only to the intended recipient	Learns which device is connected to which port by building a MAC address table . Sends frames only to the correct port — much more efficient than a hub (which broadcasts to all ports). Operates at Layer 2 (Data Link). Reduces collisions and improves network performance.
WAP (Wireless Access Point)	Allows Wi-Fi devices to join a wired network	Broadcasts an SSID (network name) that wireless devices can discover and connect to. Handles wireless encryption (WPA2/WPA3). Bridges wireless and wired segments of the network. Acts as a translator between wireless and wired protocols.
NIC (Network Interface Card)	Enables a device to connect to a network	Each NIC has a unique MAC address (48-bit hardware identifier, e.g., <code>4A:3B:2C:1D:0E:FF</code>). Can be wired (Ethernet RJ-45 port) or wireless (Wi-Fi antenna). Built into most modern devices (motherboard). Converts data from the computer into signals that can be transmitted over the network medium.

How data flows through network hardware: When you request a web page, your NIC creates a data frame with MAC addresses (for local delivery) and an IP packet with IP addresses (for routing across networks). The frame goes to your switch, which forwards it to the router based on the destination MAC address. The router strips the frame, reads the IP packet, determines the next hop from its routing table, wraps the packet in a new frame, and sends it onwards. This process repeats at each router until the packet reaches its destination.

Transmission Media

Medium	Speed	Cost	Security	Other Details
Copper Ethernet (Cat5e/Cat6/Cat6a)	Cat5e: 1 Gbps; Cat6: 10 Gbps (short runs); Cat6a: 10 Gbps (100m)	Cheap	Secure (physical access needed to tap the cable)	Reliable; limited to ~100m per run; susceptible to electromagnetic interference (EMI); easy to install
Fibre optic	Very fast (10-100+ Gbps); used for internet backbone	Expensive (cable and installation)	Very secure (extremely difficult to tap without detection; no electromagnetic emissions)	Immune to EMI; fragile (glass fibres can break); long distance (tens of km without signal loss); used for internet backbone, data centre interconnects
Wireless (Wi-Fi)	Wi-Fi 5: ~3.5 Gbps theoretical; Wi-Fi 6: ~9.6 Gbps theoretical. Actual speeds much lower.	Medium (WAP hardware)	Less secure (signals broadcast through air and can be intercepted; requires WPA2/WPA3 encryption)	Convenient; no cables; affected by walls, distance, and interference; typical indoor range 30-50m; shared medium (all devices on same channel compete for airtime)

1.3.5 The Internet, DNS and the Cloud

The internet is a global network of interconnected networks communicating via the TCP/IP protocol suite. It is the physical infrastructure (cables, routers, servers, exchanges).

The World Wide Web (WWW) is a collection of web pages and resources accessed via browsers using HTTP/HTTPS. The web is a *service that runs on* the internet — they are **not the same thing**. Other internet services include email (SMTP/IMAP), file transfer (FTP), voice calls (VoIP), video streaming, and online gaming.

DNS (Domain Name System)

The DNS converts human-readable **domain names** (e.g., `www.bbc.co.uk`) into **IP addresses** (e.g., `151.101.0.81`) that computers use to route data. Without DNS, you would need to memorise numeric IP addresses for every website. DNS is often described as the internet's phone book.

How a DNS lookup works (step by step):

1. User types `www.bbc.co.uk` in the browser.
2. The browser checks its **local DNS cache** (recently visited sites are remembered). If found, the cached IP is used immediately.
3. If not cached locally, the request goes to the **ISP's DNS server** (recursive resolver).
4. If the ISP's DNS server does not know the answer, it queries the DNS hierarchy: root server → TLD server (.co.uk) → authoritative DNS server for bbc.co.uk.
5. The authoritative DNS server returns the IP address (e.g., `151.101.0.81`).
6. The IP address is cached at each level for future requests (with a time-to-live expiry).
7. The browser uses the IP address to connect to the BBC's web server and request the page.

Web Hosting and HTTP Requests

A **web server** stores website files (HTML, CSS, JavaScript, images, databases) and serves them to clients on request. The process works like this:

1. The browser creates an **HTTP GET request** for a specific resource (e.g., `GET /index.html HTTP/1.1`).
2. This request is sent as TCP/IP packets to the web server's IP address (resolved by DNS).

3. The web server processes the request, locates the file, and sends back an **HTTP response** containing the file data and a status code (e.g., **200 OK** for success, **404 Not Found** if the page does not exist).
4. The browser receives the response and renders (displays) the web page.

The Cloud

Advantages	Disadvantages
Accessible from anywhere with an internet connection, from any device	Requires a reliable internet connection — no internet = no access to your data
Automatic backups and redundancy (data replicated across multiple data centres)	Ongoing subscription cost that accumulates over time (may exceed one-time purchase of local storage)
Scalable — easily increase storage or computing power on demand without buying hardware	Data held by a third party — privacy and security concerns. What if the provider is hacked or goes bankrupt?
No local hardware to maintain, update, or replace	Data sovereignty issues — data may be stored in another country with different laws
Facilitates collaboration — multiple users can work on the same files simultaneously	Speed limited by internet connection (uploading/downloading large files can be slow)

1.3.6 Protocols and the TCP/IP Model

A **protocol** is a set of rules that govern how data is formatted, transmitted, and received across a network. Protocols ensure that devices from different manufacturers, running different operating systems, can communicate reliably. Without agreed protocols, a Windows PC could not communicate with a Mac or a Linux server. Protocols are like languages — both sides must speak the same language to understand each other.

The TCP/IP Model (4 Layers)

The TCP/IP model divides network communication into four layers. Each layer has a specific responsibility and uses specific protocols. Data passes **down** through the layers on the sending side (each layer adds its own header — **encapsulation**) and **up** through the layers on the receiving side (each layer strips its header — **de-encapsulation**).

Layer	Name	Function	Protocols	Data Unit
4 (Top)	Application	Provides services directly to the user's application (web browsing, email, file transfer). Determines the format and meaning of the data.	HTTP, HTTPS, FTP, SMTP, IMAP, POP, DNS	Data / Message
3	Transport	Breaks data into segments; provides reliable delivery (TCP) or fast, connectionless delivery (UDP). Uses port numbers to identify which application should receive the data.	TCP (reliable, ordered, error-checked), UDP (fast, no guarantee, no order)	Segment
2	Internet	Adds source and destination IP addresses ; handles routing of packets across networks. Each router examines the destination IP and forwards the packet towards its destination.	IP (IPv4, IPv6), ICMP	Packet
1 (Bottom)	Link / Network Access	Handles physical transmission of frames over the network medium. Uses MAC addresses for local delivery within a LAN.	Ethernet, Wi-Fi (802.11)	Frame

Encapsulation example: When you send an email, the email data starts at the Application layer (SMTP). The Transport layer wraps it in a TCP segment with port numbers. The Internet layer

wraps that in an IP packet with IP addresses. The Link layer wraps that in an Ethernet frame with MAC addresses. At the receiving end, each layer strips its header in reverse order until the original email data is delivered to the email application.

Key Protocols

Protocol	Purpose	Key Detail
HTTP / HTTPS	Web browsing (transferring web pages between browser and server)	HTTPS uses SSL/TLS encryption to protect data in transit. HTTP sends data in plaintext (anyone intercepting can read it). Always look for the padlock icon and <code>https://</code> in the URL when entering sensitive information. Port 80 (HTTP) / Port 443 (HTTPS).
FTP	File transfer between computers	Upload and download files to/from a server. Uses separate control (port 21) and data (port 20) connections. Being replaced by SFTP (secure FTP) which encrypts transfers.
SMTP	Sending emails	Push protocol: pushes emails from client to mail server, and between mail servers. Port 25 (or 587 for secure). Only handles outgoing mail.
IMAP	Receiving emails (synchronised, server-side storage)	Emails stay on the server ; all devices see the same mailbox state. Read/delete on one device, change is reflected everywhere. Best for users with multiple devices. Port 993 (secure).
POP3	Receiving emails (download to local device)	Emails downloaded to the device and typically deleted from server. Only accessible from that one device. Best for single-device use or limited internet access. Port 995 (secure).
DNS	Converts domain names to IP addresses	Distributed hierarchical database of domain-to-IP mappings. Without DNS, you would need to type IP addresses. Port 53.

TCP	Reliable data delivery	Ensures data arrives complete, in order, and error-free. Uses acknowledgements and retransmission. Used for web, email, file transfers where accuracy matters.
UDP	Fast data delivery (no reliability guarantee)	No acknowledgements, no order guarantee, no retransmission. Faster than TCP. Used for live streaming, gaming, VoIP, DNS queries — where speed matters more than perfection.

1.3.7 IP and MAC Addresses

Feature	IP Address	MAC Address
Type	Logical (software-assigned, configured by network)	Physical (hardware-assigned by manufacturer, burned into NIC)
Can it change?	Yes — assigned dynamically by DHCP or manually configured; changes when connecting to different networks	No — permanently set by the manufacturer (though can be spoofed/overridden in software)
Purpose	Routes data between networks (across the internet). Used by routers.	Identifies devices within a LAN (local delivery, hop-by-hop). Used by switches.
Format	IPv4: 32 bits, written as 4 decimal octets (e.g., <code>192.168.1.1</code>). Allows ~4.3 billion addresses. IPv6: 128 bits, hexadecimal (e.g., <code>2001:0db8::1</code>). Allows 3.4×10^{38} addresses.	48 bits, written as 6 pairs of hex digits separated by colons (e.g., <code>4A:3B:2C:1D:0E:FF</code>)
Scope	Global (public IPs are unique worldwide) or local (private IPs like 192.168.x.x within a LAN)	Unique worldwide (assigned by manufacturer, in theory — though duplicates can exist in practice)

Analogy	Like a postal address (can change if you move house, identifies where to deliver)	Like a fingerprint (permanently unique to you, identifies who you are)
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IPv4 vs IPv6: IPv4 allows approximately 4.3 billion unique addresses. With billions of devices connected to the internet (phones, laptops, IoT devices, servers), we have essentially run out of IPv4 addresses. IPv6 uses 128-bit addresses, providing an astronomically larger address space (enough for trillions of devices per person on Earth). The transition from IPv4 to IPv6 is ongoing but slow because so much infrastructure still relies on IPv4.

Wired vs Wireless Connections

Feature	Wired (Ethernet/Fibre)	Wireless (Wi-Fi)
Speed	Generally faster and more consistent (dedicated cable per device)	Variable; affected by distance, interference, number of users, and obstacles
Reliability	Very reliable (dedicated physical connection)	Less reliable (signal can drop, interference from walls, microwaves, and other devices)
Security	More secure (physical access to cable needed to intercept data)	Less secure (signals broadcast through air; requires WPA2/WPA3 encryption to prevent eavesdropping)
Portability	Not portable (device must be near a cable/socket)	Portable (move freely within range of the access point)
Cost	Higher installation cost (cabling infrastructure, wall sockets)	Lower installation cost (just need a WAP; no physical cables to individual devices)
Latency	Lower and more consistent latency	Higher and more variable latency

Bluetooth: Short-range wireless technology (typically up to 10m) designed for pairing personal devices (headphones, keyboards, smartwatches, speakers). Uses less power than Wi-Fi and

operates in the 2.4 GHz band. Not designed for network access or high-bandwidth transfers. Uses frequency hopping to reduce interference.

Exam Tip: HTTP vs HTTPS: always mention that HTTPS uses **SSL/TLS encryption** to protect data in transit, while HTTP sends data in **plaintext**. DNS converts **domain names to IP addresses** — examiners want both parts stated explicitly (domain names AND IP addresses). IMAP keeps emails on the server (synced across devices); POP downloads and usually deletes from server (single device). The internet is the infrastructure; the web is one service that runs on it — they are NOT the same thing.

The TCP/IP Protocol Stack

The **TCP/IP protocol stack** is a set of communication protocols that govern how data is transmitted across the internet and other networks. It is arranged into four layers. Each layer adds its own header information as data travels down the stack (**encapsulation**) and strips it as data travels up (**de-encapsulation**). This layered approach means each layer only needs to worry about its own job — the layers above and below handle the rest.

Layer Number	Layer Name	Purpose	Protocols / Examples	Data Unit
4 (Top)	Application	Provides network services directly to user applications. Handles the meaning and format of data. This is the layer the user interacts with.	HTTP, HTTPS, FTP, SMTP, IMAP, POP3, DNS, SSH	Data / Message
3	Transport	Breaks data into segments; manages end-to-end communication. Uses port numbers to identify which application should receive data.	TCP, UDP	Segment

		TCP provides reliability; UDP provides speed.		
2	Internet (Network)	Assigns and reads IP addresses ; handles logical routing of packets across multiple networks. Each router at this layer examines the destination IP and decides the best next hop.	IP (IPv4, IPv6), ICMP	Packet
1 (Bottom)	Link (Data Link / Network Access)	Handles physical transmission of data on the local network. Uses MAC addresses for local device-to-device delivery within a single LAN segment.	Ethernet, Wi-Fi (802.11), ARP	Frame

Packet Structure

A **packet** is the fundamental unit of data transmitted across a network. Each packet has three main parts:

- **Header** — control information placed at the front of the packet. Contains:
 - **Source IP address** — where the packet came from
 - **Destination IP address** — where the packet is going
 - **Source port number** — identifies the sending application (e.g., port 443 for HTTPS)
 - **Destination port number** — identifies the receiving application
 - **Sequence number** — allows the receiver to reassemble packets in the correct order
 - **TTL (Time to Live)** — a counter decremented by 1 at each router hop; when it reaches 0, the packet is discarded, preventing it from looping forever
- **Payload** — the actual data being transmitted (a fragment of the original file, web page, email, etc.)
- **Trailer** — placed at the end; includes a **checksum** (a calculated value used to detect errors in transmission; the receiver recalculates and compares — if they differ, the packet is corrupted and must be retransmitted)

HEADER	PAYLOAD	TRAILER
(Src IP, Dst IP, Ports, Seq, TTL)	(actual data fragment)	(checksum)

TCP Three-Way Handshake

Before TCP transmits any data, it establishes a reliable connection using a **three-way handshake**. This ensures both sides are ready to communicate and agree on sequence numbers:

1. **SYN** — the client sends a synchronise (SYN) packet to the server: "I want to connect; my sequence number starts at X."
2. **SYN-ACK** — the server replies with a SYN-ACK: "Acknowledged your SYN (X+1). I want to connect too; my sequence number starts at Y."
3. **ACK** — the client sends an acknowledgement (ACK): "Acknowledged your SYN (Y+1). Connection established."

Client	Server
----- SYN (seq=X) ----->	
<----- SYN-ACK (ack=X+1) ----	
----- ACK (ack=Y+1) ----->	
Connection open	

After the handshake, data flows in both directions. TCP also uses a **four-way termination** (FIN, FIN-ACK, FIN, ACK) to close the connection cleanly.

TCP vs UDP

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented — establishes a connection via handshake before sending data	Connectionless — sends data immediately without establishing a connection first

Reliability	Reliable — acknowledgements (ACKs) confirm receipt; lost packets are retransmitted; data arrives complete and correct	Unreliable — no acknowledgements; lost packets are NOT retransmitted; some data may be lost
Ordering	Guaranteed order — sequence numbers ensure data is reassembled in the correct order	No ordering guarantee — packets may arrive out of order
Speed	Slower — overhead of handshake, ACKs, and retransmission adds latency	Faster — no handshake, no ACKs, minimal overhead
Error checking	Full error checking and correction — corrupted packets are detected and resent	Basic checksum only — corrupted packets are discarded, not resent
Use cases	Web browsing (HTTP/HTTPS), email (SMTP/IMAP), file transfer (FTP), online banking — anywhere data accuracy is critical	Live video streaming, online gaming, VoIP (video calls), DNS queries — where speed matters more than perfection and a dropped frame is acceptable
Analogy	A tracked delivery service — you get confirmation the parcel arrived; if it is lost it is resent	Broadcasting on radio — listeners receive what they can; there is no way to request a retransmission of a missed word

Exam Tip: TCP vs UDP: always tie your answer to a specific use case. TCP is used where every byte must arrive correctly (file download, web page, email). UDP is used where real-time delivery matters more than perfection (live stream, gaming, VoIP) — a dropped frame in a video call is better than pausing to wait for retransmission. Examiners want you to justify, not just identify.

Network Hardware — Detailed Comparison

Network hardware devices operate at different layers of the network stack (the layer numbers below follow the 7-layer OSI convention: Layer 1 physical, Layer 2 data link, Layer 3 network, not the 1-4 numbering of the TCP/IP table above). Understanding **which layer each device operates at** and **what address type it uses** is essential for explaining how data moves through a network.

Device	Layer	Address Used	Purpose and Key Details
NIC (Network Interface Card)	Link (Layer 1)	MAC address (burned in by manufacturer)	Every device needs a NIC to connect to a network. Each NIC has a globally unique 48-bit MAC address (e.g., A4:C3:F0:85:AC:2D) assigned at manufacture and stored in hardware. Can be wired (RJ-45 Ethernet socket) or wireless (Wi-Fi antenna). Converts digital data from the CPU into electrical, optical, or radio signals for transmission.
Hub	Link (Layer 1)	None (dumb device)	A dumb repeater that broadcasts every incoming signal to all ports simultaneously. All connected devices receive every packet, even those not addressed to them. Creates a single collision domain — if two devices send at the same time, the signals collide and both must retransmit. Generates unnecessary network traffic. Now obsolete — replaced by switches in all modern networks.

Switch	Data Link (Layer 2)	MAC address	An intelligent device that learns which device is connected to which port by building a MAC address table . When a frame arrives, the switch looks up the destination MAC address and sends it only to the correct port . This eliminates unnecessary traffic, reduces collisions, and significantly improves network efficiency. Each port is its own collision domain. Standard device for wired LANs.
Router	Internet (Layer 3)	IP address	Connects different networks together (e.g., your home LAN to the internet). Reads the destination IP address in each packet and uses a routing table to determine the best next hop. Performs NAT (Network Address Translation) to map multiple private IP addresses to a single public IP. Also typically includes a built-in DHCP server, firewall, and WAP in home models.
WAP (Wireless Access Point)	Link (Layer 1/2)	MAC address (wireless)	Bridges wireless devices onto a wired LAN. Broadcasts an SSID (network name) that devices can see and connect to. Handles wireless encryption (WPA2/WPA3) to protect the wireless segment. A home router typically has a built-in WAP. Enterprise networks use multiple WAPs to provide full coverage across a building.
Modem	Link (Layer 1)	N/A	Modulates digital signals from a computer into analogue signals suitable for transmission over a telephone line (ADSL) or cable, and demodulates incoming analogue signals back to digital. The word modem comes from <i>modulator-demodulator</i> . Required to connect to the internet via a telephone line. Modern fibre connections use an ONT (Optical Network Terminal) instead, as fibre carries digital signals natively.

Hub vs Switch vs Router — Key Distinctions

This comparison is one of the most frequently examined topics in the networks unit. Make sure you can clearly explain the difference between all three:

	Hub	Switch	Router
What it connects	Devices within a LAN	Devices within a LAN	Different networks (LAN to WAN, LAN to internet)
Address used	None — broadcasts to all ports regardless	MAC address — sends to correct port only	IP address — routes between networks
Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Internet/Network)
Intelligence	None (dumb device)	Builds MAC address table; learns port locations	Builds routing table; calculates best path
Traffic generated	High — broadcasts all data to all ports	Low — sends data only to the intended recipient	Routes packets between networks efficiently
Collisions	One large collision domain — frequent collisions	Each port is a separate collision domain — no collisions between ports	Operates at a higher level — not concerned with collisions
Status	Obsolete — no longer used in modern networks	Standard LAN device	Essential for internet connectivity

Exam Tip: Hub vs switch: a hub broadcasts to ALL ports; a switch sends only to the correct port using its MAC address table. In an exam, always say "reduces unnecessary network traffic" and "reduces collisions" as advantages of a switch over a hub. This single distinction accounts for a large share of marks in network hardware questions.

Exam Tip: Know that MAC addresses are hardware addresses (fixed, burned into NIC, Layer 2) and IP addresses are logical addresses (assigned by DHCP or manually configured, Layer 3). A switch uses MAC addresses to forward frames within a LAN; a router uses IP addresses to route packets between different networks. Switches do not read IP addresses; routers do not typically build MAC address tables.

IP and MAC Addressing — In Depth

IP Addresses

An **IP address** (Internet Protocol address) is a **logical address** assigned to a device on a network. It identifies the device's location on the network and is used by routers to route packets to the correct destination. IP addresses can change — they are assigned by DHCP or configured manually.

	IPv4	IPv6
Bit length	32 bits	128 bits
Format	Dotted decimal — four octets (0–255) separated by dots, e.g., 192.168.1.1	Hexadecimal — eight groups of four hex digits separated by colons, e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334 (can be shortened by omitting leading zeros)
Address space	Approximately 4.3 billion unique addresses — now exhausted	3.4×10^{38} unique addresses — effectively limitless
Why needed?	Original internet addressing system	Introduced because IPv4 addresses have run out due to the explosion of connected devices (smartphones, IoT, etc.)

Private vs Public IP Addresses

	Private IP Address	Public IP Address
Used where?	Within a LAN (home, office, school) — not routable on the public internet	On the internet — visible to the outside world
Assigned by	Router's DHCP server automatically assigns from reserved private ranges	Assigned by ISP
Ranges (IPv4)	192.168.x.x , 10.x.x.x , 172.16.x.x – 172.31.x.x	All other ranges
Uniqueness	Only unique within the local network — the same private IP can exist in millions of different LANs simultaneously	Globally unique — no two devices on the internet share the same public IP at the same time
Example	Your laptop at home: 192.168.1.105	Your router's internet-facing address: 82.34.167.9

DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns network configuration to devices when they join a network. Without DHCP, every device would need to be manually configured with an IP address — impractical in large networks or for guests.

When a device joins a network, it broadcasts a DHCP request. The DHCP server (usually the router) responds with:

- **IP address** — a unique address for the device on this network (e.g., 192.168.1.15)
- **Subnet mask** — defines which part of the IP address identifies the network vs the host (e.g., 255.255.255.0)
- **Default gateway** — the router's IP address (used when sending data outside the LAN)
- **DNS server address** — the address of the DNS server to use for name resolution

The IP address is leased for a set time. When the lease expires, the device must request a renewal — this is why your device's IP address can occasionally change.

DNS — Domain Name System (Detailed)

DNS is a hierarchical, distributed database that translates human-readable domain names into IP addresses. The hierarchy works as follows:

1. **Local cache** — the device checks if it has recently resolved the domain and cached the result. If found, the cached IP is used immediately (fastest).
2. **ISP's recursive resolver** — if not cached locally, the query goes to the ISP's DNS server, which searches its own cache.
3. **Root servers** — if the ISP's server does not know, it queries one of the 13 root server clusters, which know the addresses of TLD servers.
4. **TLD (Top-Level Domain) server** — the root server directs the query to the TLD server for the domain suffix (e.g., `.com` , `.co.uk` , `.org`).
5. **Authoritative name server** — the TLD server points to the authoritative server for the specific domain (e.g., `google.com`), which holds the definitive IP address record and returns it.

Each result is cached at every level with a **TTL (Time to Live)** value — this is how long it should be kept before re-querying. Caching dramatically speeds up future lookups for popular sites.

NAT — Network Address Translation

NAT allows a router to map many private IP addresses (inside the LAN) to a single public IP address (facing the internet). This conserves the IPv4 address space and provides a basic layer of security (internal devices are not directly accessible from the internet).

When a device inside the LAN sends a request to the internet: the router replaces the private source IP with its public IP and records the mapping. When the response arrives, the router looks up the mapping and forwards the packet to the correct internal device. From the internet's perspective, all traffic appears to come from one public IP address.

Exam Tip: DHCP automatically assigns IP address, subnet mask, default gateway, and DNS server — know all four items. DNS translates domain names to IP addresses — examiners

want both parts stated. NAT allows multiple devices to share one public IP address — it extends the usable life of IPv4 and provides a degree of security by hiding internal IP addresses from the internet.

Wireless Networks

Wi-Fi Standards

Standard	Common Name	Frequency	Max Theoretical Speed	Notes
802.11a	Wi-Fi (legacy)	5 GHz	54 Mbps	Early standard; limited range; now obsolete
802.11b	Wi-Fi (legacy)	2.4 GHz	11 Mbps	Slow; widely compatible; now obsolete
802.11g	Wi-Fi (legacy)	2.4 GHz	54 Mbps	Improved speed over 802.11b; still used in older devices
802.11n	Wi-Fi 4	2.4 GHz & 5 GHz	600 Mbps	Dual-band; MIMO antennas; still common in homes
802.11ac	Wi-Fi 5	5 GHz (primarily)	3.5 Gbps	MU-MIMO; beamforming; dominant in current devices
802.11ax	Wi-Fi 6	2.4 GHz & 5 GHz	9.6 Gbps	Improved performance in dense environments; OFDMA; lower latency; better battery efficiency for devices

SSID (Service Set Identifier) is the name of a wireless network that devices can discover and connect to. When you see a list of available Wi-Fi networks on your phone, each name is

an SSID. SSIDs can be **broadcast** (visible to everyone nearby — easier to connect) or **hidden** (the network name is not advertised — devices must know the SSID in advance to connect). Hiding the SSID provides minimal security — network scanning tools can still detect it.

2.4 GHz vs 5 GHz

Feature	2.4 GHz	5 GHz
Range	Greater range — lower frequency waves travel further and penetrate walls and obstacles better	Shorter range — higher frequency waves lose energy more quickly and are blocked more by walls
Speed	Slower maximum speeds	Faster maximum speeds
Congestion	More congested — only a few non-overlapping channels; shared with microwaves, Bluetooth, baby monitors, and many older devices	Less congested — more available channels; fewer competing devices
Best for	Devices far from the router; IoT devices that don't need high speed; devices in adjacent rooms	Devices close to the router requiring high speeds; streaming 4K video; gaming

Wireless Security

- **WPA2 (Wi-Fi Protected Access 2)** — uses AES-CCMP encryption (128-bit key). The current standard for most home and business Wi-Fi. Replaced the weaker WEP and WPA standards. Can be vulnerable to offline dictionary attacks against weak passwords.
- **WPA3** — the latest standard, introduced in 2018. Uses SAE (Simultaneous Authentication of Equals) handshake instead of PSK, making offline dictionary attacks much harder. Forward secrecy means past sessions cannot be decrypted even if the password is later compromised. Increasingly required for new Wi-Fi certified devices.

- **WEP (Wired Equivalent Privacy)** — obsolete; broken encryption; must not be used. Can be cracked in minutes.

Ethernet (Wired) vs Wi-Fi (Wireless)

Feature	Ethernet (Wired)	Wi-Fi (Wireless)
Speed	Generally faster and consistent — dedicated connection; modern Ethernet supports 1–10 Gbps	Variable — affected by distance, walls, interference, and number of users sharing the channel
Reliability	Very reliable — physical connection does not suffer from interference or signal drop	Less reliable — signal can be disrupted by walls, microwave ovens, other Wi-Fi networks, and interference
Security	More secure — attacker needs physical access to the cable or network hardware to intercept data	Less secure — signals broadcast through the air and can be intercepted if encryption is weak or absent; requires WPA2/WPA3
Latency	Lower and more consistent — important for gaming and real-time applications	Higher and more variable — wireless contention and signal processing add delay
Convenience	Device must be physically near a cable socket — limits mobility	Allows full mobility within the coverage area — ideal for laptops, phones, tablets
Installation cost	Higher — requires cabling infrastructure, wall sockets, and switches throughout the building	Lower — only requires a WAP; no cables to individual devices
Best for	Desktop computers, games consoles, servers, anywhere consistent speed and reliability matter	Mobile devices, laptops, smartphones, anywhere portability matters

Exam Tip: When comparing wired and wireless, always address at least four of: speed, reliability, security, latency, convenience, and cost. Examiners expect comparisons, not just descriptions — always say "wired is faster than wireless because..." rather than just "wired is fast." For wireless security questions, name the protocol: WPA2 uses AES encryption; WPA3 improves resistance to offline attacks using SAE.

Network Protocols — Reference Table

A **protocol** is a set of agreed rules that govern how data is formatted, transmitted, received, and acknowledged between devices. Protocols allow devices from different manufacturers running different software to communicate reliably. **Port numbers** identify which application or service a packet is destined for — like a door number in a building. The IP address gets the packet to the building (device); the port number directs it to the right room (application).

Protocol	Full Name	Port(s)	What It Does	Key Point
HTTP	HyperText Transfer Protocol	80	Transfers web pages (HTML, CSS, images) between a web server and browser	Data sent in plaintext — anyone who intercepts traffic can read it. Not suitable for sensitive information.
HTTPS	HyperText Transfer Protocol Secure	443	HTTP wrapped in TLS/SSL encryption — secure web browsing	Data is encrypted end-to-end. Uses a digital certificate from a Certificate Authority (CA) to verify the server is genuine. Always use HTTPS for banking, shopping, or login pages. Shown by padlock in browser.

FTP	File Transfer Protocol	20 (data), 21 (control)	Transfers files between a client and a server over a network	Uses two separate connections: port 21 for commands (control), port 20 for actual file data. Sends data in plaintext — SFTP (port 22) or FTPS encrypt the transfer.
SMTP	Simple Mail Transfer Protocol	25 (587 secure)	Sends outgoing email from client to mail server, and between mail servers	Push protocol — pushes emails out. Only handles <i>sending</i> ; you need IMAP or POP3 to <i>receive</i> email. Think of SMTP as the postal worker who delivers letters.
IMAP	Internet Message Access Protocol	143 (993 secure)	Retrieves emails from the server; emails remain stored server-side	Emails stay on the server — all devices see the same mailbox. Read, delete, or move on one device and it is reflected everywhere. Best for users with multiple devices (phone, laptop, tablet).
POP3	Post Office Protocol version 3	110 (995 secure)	Downloads emails to the local device and typically deletes them from server	Emails downloaded to one device — not accessible from other devices. Useful where internet access is unreliable (work offline) or server storage is limited. Best for single-device use.

SSH	Secure Shell	22	Provides encrypted remote command-line access to another computer	Replaced Telnet (which sent commands in plaintext). Used by system administrators to manage servers remotely. All traffic is encrypted — commands and responses cannot be intercepted.
DHCP	Dynamic Host Configuration Protocol	67 (server), 68 (client)	Automatically assigns IP address, subnet mask, gateway, and DNS server to devices joining a network	Without DHCP, every device must be manually configured. DHCP assigns a lease — the IP is loaned for a set time and may change on renewal.
DNS	Domain Name System	53	Translates human-readable domain names (e.g., <code>www.bbc.co.uk</code>) into IP addresses (e.g., <code>151.101.0.81</code>)	Without DNS, you would need to memorise IP addresses for every website. DNS is often called the "phone book of the internet." Queries use UDP by default; TCP for large responses.

Exam Tip: Know port numbers for: HTTP (80), HTTPS (443), FTP (20/21), SMTP (25), IMAP (143), POP3 (110), SSH (22), DNS (53). The most common exam question is IMAP vs POP3: IMAP stores emails on the server (syncs across all devices); POP3 downloads emails to one device and removes them from the server. HTTPS vs HTTP: HTTPS uses TLS/SSL encryption — always say "encrypted" and mention the certificate authority or padlock icon.